

SANGJIN LEE
New York, United States
+82-10-93299427 | lee.sangjin@stonybrook.edu | [LinkedIn](#)

April 27, 2026

Dr. J. Nicholas Cochran

<https://www.linkedin.com/in/j-nicholas-nick-cochran-34445675/>

Dear Dr. J. Nicholas Cochran,

Turning data into insights has been my motto throughout my time as an undergraduate student. Contributing to genomic discoveries for human health, I have been deeply interested in this field for my whole life. HudsonAlpha's philosophy and approach to innovation have led me to set my goal as a Computational Biologist at the Cochran Lab. Thus, the Computational Biologist position you advertised will be a great opportunity for me. I have enclosed my resume for your consideration.

I am majoring in Applied Mathematics and Statistics and minoring in Computer Science at Stony Brook University, New York, and expect to graduate in the spring of 2027. I am fluent in both English and Korean, which is useful for communicating with Koreans and Americans. I have a strong understanding of biotechnology industry, including a deep understanding of the HudsonAlpha Institute. Also, I have worked at the Korea Polar Research Institute as a Student Researcher. While working, I have covered diverse tasks, including molecular dynamics simulations in GROMACS, conducting research and analyzing the correlation between liquid flow velocity and protein binding degrees to establish functional relationships. Modeling mathematics to develop functions for flow velocity variation as a function of pipe diameter to optimize experimental conditions and improve fluid dynamics efficiency.

Also, throughout my academic journey, I have worked as a Resident Advisor, hosting and managing diverse events, and as a Teaching Assistant, supervising more than 200 students. Though I have participated in diverse extracurricular activities, I have maintained a GPA of 3.86, deepening my understanding of academics. I believe these experiences have prepared me to work as a Computational Biologist at the HudsonAlpha Institute for Biotechnology. I would appreciate the opportunity to discuss my qualifications more fully in an interview. Thank you for your consideration.

Sincerely,

Sangjin Lee

About the job

(https://www.linkedin.com/jobs/view/4420222323/?alternateChannel=search&eBP=NOT_ELIGIBLE_FOR_CHARGING&refId=6AVRr3fk%2FJLnsjSgV%2BM%2FYw%3D%3D&trackingId=Lj%2Ba8dJTDgbnh1Qdq%2Fyt4Q%3D%3D)

Position: Computational Biologist I - Cochran Lab

Job Type: Full-Time

Institute Overview

The HudsonAlpha Institute for Biotechnology (www.hudsonalpha.org) is a non-profit organization with the mission to use biotechnology and basic biological research to make important discoveries, improve human health, stimulate economic development, inspire young people to seek careers in science and teach members of the public the importance of science in their everyday lives. Located in Cummings Research Park in Huntsville, Alabama, one of the world's leading science and technology parks, HudsonAlpha's campus contains state-of-the-art laboratories for biological research and development, as well as education, in the areas of genomics, genetics and personalized medicine.

Department Overview

The J. Nicholas Cochran laboratory uses genomic approaches to answer biological questions specific to neurodegenerative diseases with an emphasis on Alzheimer's disease and other dementias. Therefore, while not required, subject area familiarity and/or interest in genetics, genomics, neuroscience, or neurodegenerative diseases is desirable. Projects include the study of gene regulation for genes implicated in neurodegeneration, novel analysis approaches for associating genetic variation with neurodegenerative diseases, and genome-wide genomics approaches relevant to neurodegeneration. Key ongoing projects related to this role include several national and international collaborations on genetic contributors to neurodegenerative diseases, including with the Multi-Partner Consortium to Expand Dementia Research in Latin America (ReDLat), the UCSF Memory and Aging Center, and the UAB Alzheimer's Disease Research Center.

Position Summary

The Computational Biologist I combines computer science, statistics, mathematics, and engineering to analyze and interpret biological data. This position supports collaborative research efforts with experimental scientists, software engineers, and other bioinformaticians to implement computational workflows and data analysis strategies.

The Computational Biologist I uses lab-generated or publicly available data to answer specific biological questions. By applying computational methods, the Computational Biologist I is integral to processing and analyzing the large amount of data generated in or accessed by a genomics laboratory. The ideal candidate will be comfortable constructing bioinformatics

pipelines that interface with internal and external resources using modern programming languages commonly employed in bioinformatics (Python, R, and bash), both locally and in high-performance computing (HPC) and cloud environments.

JOB SPECIFIC TASKS

Data Analysis & Research

- Perform data preprocessing, quality control, and analysis using established computational workflows
- Understand the biological science behind the laboratory projects
- Analyze genomic and transcriptomic data from neurodegenerative disease studies
- Process high-throughput sequencing data (RNA-seq, DNA-seq, ATAC-seq, single-cell sequencing)

Manage access, management, and processing for large biological datasets

•

Pipeline Development & Maintenance

- Assist in maintaining bioinformatics tools and pipelines to support research initiatives
- Construct and optimize bioinformatics workflows for genomic data analysis
- Collaborate with senior team members to troubleshoot data-related challenges and optimize workflows
- Implement version control practices (Git/GitHub) for collaborative code development

Work with workflow management systems (Snakemake, Nextflow, WDL) as appropriate

•

Collaboration & Communication

- Collaborate with the research team to plan and execute computational applications
- Communicate technical findings clearly across teams and document methods for reproducibility
- Work with experimental biologists, clinicians, and other computational researchers
- Present analyses at lab meetings and research forums

Contribute to peer-reviewed research papers

•

Education

EDUCATION AND REQUIRED SKILLS EXPERIENCE

- Bachelor's degree or master's degree in computational biology, bioinformatics, biostatistics, computer science, statistics, or related quantitative field

Master's degree preferred

-

Experience

- 0 to 3 years of experience in bioinformatics, computational biology, or related field
Experience may include postgraduate research, internships, industry positions, or significant coursework projects

-

Required Technical Skills

- Proficiency in Python and/or R and bash scripting
 - Experience with version control systems (Git/GitHub)
 - Strong understanding of bioinformatics principles and proficiency in relevant computational tools
 - Fluency with Unix/Linux command-line environments
 - Ability to function in a high-performance computing (HPC) cluster environment
 - Strong understanding of statistics and quantitative data analysis
- Proficient with Google Workspace and collaborative tools (Zoom, Slack, etc.)

-

Compensation & Benefits

Salary Range: \$66,000 - \$75,000 annually, commensurate with education and experience.

-

Benefits Include

- Comprehensive health, dental, and vision insurance
 - Retirement plan with employer contribution
 - Generous paid time off
 - Professional development support
- Access to cutting-edge computational resources and technologies

-

Physical Demands/Environmental Factors/Mental Functions

ADA Requirements – The physical demands, work environment factors, and mental functions described below are representative of those that must be met by an employee to successfully perform the essential function of this job. Reasonable accommodations may be made to enable individuals with disabilities to perform the essential functions.

Disclaimer - All job requirements are subject to possible revision to reflect changes in the position requirements or to reasonably accommodate individuals with disabilities. Some

requirements may exclude individuals who pose a threat or risk to the health and safety of themselves or other employees. This job description in no way states or implies that these are the only duties which will be required in this position. Employees will be required to follow other job-related duties as requested by their supervisor/manager (within guidelines and compliance with Federal and State Laws). Continued employment remains on an "at-will" basis.

Physical Demands

- Must be able to remain in a stationary position (sit or stand) for prolonged periods (50% of the time).
- Frequently moving between work areas.
- Frequently moving, transporting, and positioning materials up to 30 pounds.
- Vision abilities required by this job include close vision, distance vision, and depth perception.
- Frequent operation of common office equipment (e.g. computers, mobile devices, calculators, copiers, scanners) and laboratory equipment.

Frequently position self (e.g. stoop, crouch, crawl, bend) and pull/push, grasp/squeeze, wrist flexion/extension, reach to operate and maintain laboratory equipment.

•

Environmental Factors

- Exposure to chemical, mechanical and electrical hazards, and hazardous physical substances and fumes, and airborne particles.

Noise level in the work environment is usually moderate.

•

Mental Functions

- Continuous mental and visual attention, requiring sustained attention and observation.
- Excellent fine motor skills and hand control in order to pipette accurately.
- Ability to sustain focus during research procedures (to maintain protocol such as order of operations).
- Ability to visually observe results and record outcomes on lab logs and/or PC.
- Basic computer skills are required.
- Extensive documentation skills, good attention to detail, and exceptional organizational skills are required.
- Communicate effectively, both orally and in writing, in the English language.
- Requires the ability to adapt to change efficiently.
- Ability to independently lead and conduct research, and to evaluate, interpret and present complex scientific data.
- Capacity to work productively within functionally diverse teams as well as collaborate with external investigators and partners in academic and industrial settings.
- Ability to prioritize issues and use sound judgment to recognize when additional consultation and/or help is needed.

- Comfortable working on multiple projects at one time.
- Strong logical thinking and deduction skills.
- Ability to effectively present ideas and information in a variety of formats to a wide range of audiences.

Positive, flexible, and professional attitude.

-

It is the policy of HudsonAlpha Institute for Biotechnology, in accordance with all applicable laws, to recruit, hire, train, and promote persons in all job titles without regard to race, color, religion, sex, age, national origin, disability, genetic information, sexual orientation, veteran status, or any other basis prohibited by applicable law.